Aiden Tran

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Education

University of California, Berkeley 2029 - Berkeley, CA

BA in Computer Science

- **Relevant Coursework:** Structure/Interpretation of Computer Programs, Data Structures, Discrete Math/Probability Theory, Linear Algebra, Computational Astrophysics

Projects

ML Library & LLM Inference Engine | *C++, AI/ML, CMake* GitHub

- Built a C++17 machine learning library from scratch, covering SVM, linear and logistic regression, KNN, decision trees, and random forest, all without an external ML framework, to understand how the algorithms work without relying on black box implementations.
- Implemented a Llama 3.2-1B inference stack: GGUF weight dequantization, tokenization, multi-head attention, and KV caching, then added a CUDA path that runs generation 42x faster than my own CPU only version.
- Wrote AVX2 SIMD vectorization by hand for matrix multiplication, cutting training and inference time by up to 2.5x on large datasets.
- Implemented backpropagation and SMO based SVM training by hand, including a precomputed kernel cache that turns repeated per-pair kernel evaluations into O(1) lookups instead of recomputing them on every SMO iteration.

Multiplayer Fighting Game | *Lua, Networking*

- Developed a competitive 2.5D fighting game featuring industry-standard rollback networking capable of resimulating 12 frames in less than 3ms, well under the 16.6ms frame budget at 60fps.
- Designed a scalable object-oriented architecture to manage a deterministic game loop and 60+ unique states per entity.
- **Shipped to a live player base and accumulated approximately 500k impressions**, collecting user feedback throughout development
- Published a technical writeup on the rollback networking implementation (Link to Writeup), explaining concepts, implementation, and challenges faced during development

MarkovLuau | *Lua, Procedural Generation* GitHub

- Built a Lua extension of MarkovJunior (Source), giving developers a headless library of Markov chain rewrite and Wave Function Collapse algorithms to easily generate dynamic, rule-based procedural content
- Designed a programmatic DSL for writing rule sequences directly in code, alongside a parser that runs the original MarkovJunior XML models unmodified
- Built an interactive visualization tool, allowing users to preview generated outputs step-by-step and debug rule sequences
- Wrote complete documentation covering installation, getting started, worked examples, and a full API reference (Link to Documentation)

Real-Time Audio Signal Processor | *C++, ImGui* GitHub

- Developed a C++ desktop application that captures live audio and analyzes frequencies in real time using a Cooley-Tukey FFT pipeline, including framing, windowing, and peak detection to identify and map fundamental frequencies to musical notes
- Rendered real-time visual feedback including waveform graphs, frequency spectrums, and detected note indicators using ImGui for a responsive and interactive UI
- Supported both real-time microphone input and static audio file analysis, processing imported recordings and returning the correlating musical note for each segment

Black Hole Spacetime Analyzer | *Python, NumPy, Matplotlib, Pandas* GitHub

- Built a Python simulation modeling particle propagation through curved spacetime near black holes, implementing numerical integration of geodesic paths using the Schwarzschild metric to compute gravitational time dilation and redshift effects
- Integrated the AGN Black Hole Mass Database to run simulations against real supermassive black hole data, validating the model against observed astrophysical parameters
- Developed an interactive parameter system allowing users to adjust mass, radius, and observer frame and observe their impact on spacetime in real time
- Built a 3D visualization engine in Matplotlib rendering spacetime curvature surfaces and geodesic paths, graphically demonstrating event horizon boundaries for infalling objects

Experience

Key On Harmony Jul. 2024 – Aug. 2024

Software Engineering Intern

- Developed a secure desktop ingestion client using Electron and React, replacing manual entry with a schema-validated bulk upload system

Technical Skills

Languages: C++, Python, Java, C#, Lua, TypeScript, SQL
Frameworks/Technologies: CMake, React, Node.js, Next.js, PostgreSQL, Unity, Godot
Tools & Infrastructure: Git, Docker, Linux